

15.4 部分习题参考简答

1. (1) $\frac{12\pi R^5}{5}$. (2) $4\pi abc$.
2. 2π .
4. (1) 是保守场, 势函数为 $\sin(xy) - \cos z$, $\int_{(4,0,1)}^{(2,1,-1)} \mathbf{V} \cdot d\mathbf{r} = \sin 2$;
(2) 非保守场.
5. $\lambda = -\frac{1}{2}$.
6. (1) $\mathbf{0}$; (2) $(xz, -yz, 0)$; (3) $\mathbf{0}$.
7. (1) $2yz + 2zx + 2xy$; (2) $y + ze^{yz} + x \cos(zx)$; (3) $\mathbf{0}$.
8. 势函数 $u = \frac{1}{2}x^2y^2 + 2xy - y^2 + \frac{1}{2}z^2 + C$, $\forall C \in \mathbb{R}$.
9. $\text{grad } f = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}}(x, y, z)$;
 $\text{div}(f(r)\mathbf{r}) = 3f(\sqrt{x^2 + y^2 + z^2}) + \sqrt{x^2 + y^2 + z^2} f'(\sqrt{x^2 + y^2 + z^2})$;
 $\text{rot}(f(r)\mathbf{r}) = \mathbf{0}$.